Notes: Ha, Ma, and Žaldokas (JFE, 2024)

Motivating Collusion

Contents

1 Big picture

- **Question.** Can CEO compensation be designed to make managers more willing to collude with product-market peers when antitrust enforcement weakens?
- **Core idea.** Shareholders may benefit from softer product-market competition, but CEOs personally bear legal, career, and reputational costs from explicit collusion. Compensation contracts can partially close this wedge.
- **Empirical shock.** The paper studies the 2013 closure of four U.S. Department of Justice Antitrust Division regional offices: Atlanta, Cleveland, Dallas, and Philadelphia. The reform increased the distance between many firms and their local antitrust enforcement office.
- **Treatment.** The main treatment is $\Delta Distance_i$, measured in 100-mile units, equal to the increase in geographic distance between a firm's headquarters and its covering DoJ antitrust office after the closures.
- **Mechanism.** If collusion becomes more attractive to shareholders, boards can weaken executives' incentives to outperform local rivals. In pay-performance regressions, this means CEO pay should become more positively related to local peer returns and less strongly related to own-firm returns.
- **Main outcome.** The key outcome is the sensitivity of CEO compensation to the stock returns of the firm and of local product-market peers.
- **Main result.** After the DoJ closures, firms more exposed to the reform increased the loading of CEO pay on local peer performance. In the authors' preferred local-peer specification, a 100-mile increase in distance raises the local-peer pay sensitivity by about 0.018.
- **Own-return result.** The same specification shows a negative change in own-return pay sensitivity of about -0.016, consistent with weaker incentives to compete aggressively.
- **Compensation channel.** The effect is stronger for cash compensation than for equity compensation, suggesting active contract redesign rather than a mechanical equity-value effect.
- **Heterogeneity.** Effects are stronger when collusion is more plausible or more valuable: strategic complements, concentrated industries, industries with more public firms, geographically concentrated operations, better shareholder-oriented boards, older CEOs, and more mobile executive labor markets.
- **Product-market outcomes.** Exposed firms show higher gross profit margins after the reform, and firms/industries with stronger compensation changes show higher profitability and higher return comovement with local peers.
- **Other contract features.** Affected firms lengthen equity vesting horizons and shift performance metrics toward strategic goals and profit margins, while reducing sales-based metrics.
- **Policy takeaway.** Corporate-governance alignment can conflict with consumer welfare. When antitrust enforcement weakens, shareholder-friendly compensation design may make collusion easier.

2 Data and measurement

2.1 Sample and main datasets

- The main sample covers U.S. publicly listed firms from 2008 to 2017.
- The focus is on CEO compensation because cartel coordination is usually a senior-management decision, and top-level incentives can cascade through the organization.
- CEO compensation data come from ExecuComp.
- Compensation variables include total compensation, cash compensation, stock and option awards, and equity vesting information.
- Explicit benchmark and vesting-contract details come from Incentive Lab.
- Stock returns come from CRSP; financial controls come from Compustat.
- Product-market peers are based on Hoberg and Phillips text-based product similarity.
- Convicted cartel cases are from Connor's cartel database.
- State-level economic statistics come from the Census Bureau.
- Historical headquarters are collected from SEC filings.
- DoJ antitrust office jurisdictions come from the Antitrust Division and related institutional records.

Interpretation: The paper combines corporate-finance data, antitrust-enforcement geography, and product-market peer networks. The empirical design uses the DoJ reform to shift the expected cost of local collusion while observing how boards adjust CEO contracts.

2.2 Treatment: exposure to DoJ office closures

The 2013 reform closed the DoJ Antitrust Division field offices in Atlanta, Cleveland, Dallas, and Philadelphia. Some casework was moved to remaining regional offices and Washington, DC. The authors define exposure as

$$\Delta Distance_i = \frac{Distance_i^{post} - Distance_i^{pre}}{100}, \tag{1}$$

where distance is from firm headquarters to the covering DoJ field office, and the variable is measured in 100-mile units.

- $Post_t = 1$ for years 2013 and after.
- $\Delta Distance_i = 0$ for firms whose covering office did not change or whose new office was not farther away.
- The continuous treatment captures intensity: the farther the new antitrust office, the weaker local monitoring is expected to be.
- Binary-treatment robustness checks use cutoffs such as $\Delta Distance > 0$ and $\Delta Distance > 400$ miles.

Interpretation: The identification relies on the idea that local enforcement is more effective when prosecutors are geographically close to firms and local information sources. Moving a firm's covering office farther away weakens "boots on the ground" antitrust monitoring.

2.3 Peers and local product markets

The authors define local product-market peers as firms satisfying both conditions:

1. They are among the focal firm's Hoberg-Phillips product-market peers, with similarity scores in the top 70%.
2. They are headquartered within a 200-mile radius of the focal firm.

- Local peers are likely to face common local shocks.
- Local peers are also more plausible collusion partners than geographically distant product-market rivals.
- Non-local peers help separate local collusion effects from broad industry performance shocks.

Interpretation: This is central to the paper. Standard relative performance evaluation predicts CEO pay should load negatively on peer returns to filter common shocks. The collusion channel predicts the opposite for local product-market peers after enforcement weakens.

2.4 Compensation outcomes

The paper studies several compensation variables:

- $\ln(1 + \text{Total compensation})$.
- $\ln(1 + \text{Cash compensation})$.
- $\ln(1 + \text{Equity compensation})$.
- Vesting horizon of time-vesting equity plans.
- Explicit RPE adoption and RPE peer-group composition.
- Use of strategic, profit-margin, and sales performance metrics in incentive plans.

Interpretation: The authors first test implicit pay-performance sensitivity. They then examine whether explicit contract provisions move in the same direction: less direct competitive benchmarking, more strategic/profit-margin goals, longer vesting, and less sales emphasis.

2.5 Product-market outcomes

The product-market outcomes are:

- Gross profit margin, used as an observable proxy for markups or pricing power.
- Return comovement with local peers, measured as the average annual correlation of weekly stock returns between the focal firm and its local peers.

Interpretation: If compensation changes support collusion, affected firms should not only change contracts; they should also look more profitable and more coordinated with local rivals.

3 Empirical specifications and interpretation

3.1 Baseline pay-performance regression

The main difference-in-differences specification is

$$\ln(Compensation_{i,t}) = \beta_1 \Delta Distance_i \times Post_t \times \ln(Return_{i,t}) \tag{2}$$
$$+ \beta_2 \Delta Distance_i \times Post_t \times \ln((Local)PeerReturn_{i,t}) \tag{3}$$
$$+ K_{i,t} + X_{i,t} + \tau_t + \gamma_i + \varepsilon_{i,t}. \tag{4}$$

- $\ln(Compensation_{i,t})$ is total, cash, or equity CEO compensation.
- $\ln(Return_{i,t})$ is own-firm stock return.
- $\ln((Local)PeerReturn_{i,t})$ is the average stock return of product-market peers, especially local peers.
- $K_{i,t}$ includes lower-order terms and interactions required by the triple interaction design.
- $X_{i,t}$ includes controls such as size, sales growth, and CEO tenure.
- τ_t are year or industry-year fixed effects.
- γ_i are firm fixed effects.

Interpretation: $\beta_2 > 0$ means that after local antitrust monitoring weakens, exposed firms reward CEOs more for local peers doing well. This is consistent with discouraging aggressive competition. $\beta_1 < 0$ means the same firms weaken own-firm performance sensitivity, also consistent with less emphasis on beating rivals.

3.2 Cash versus equity compensation

The same specification is estimated separately for

$$Y_{i,t} \in \{\ln(1 + CashComp_{i,t}),\ \ln(1 + EquityComp_{i,t})\}. \tag{5}$$

Interpretation: If the effect is driven by active contract redesign, the more flexible component—cash compensation—should respond more strongly. If the effect is mechanical, equity compensation should move more because stock grants mechanically rise when stock values rise. The paper finds the former.

3.3 Robustness using binary treatment

The continuous exposure measure is replaced with threshold indicators:

$$D(\Delta Distance_i > x), \qquad x \in [0, 1000] \text{ miles}. \tag{6}$$

Interpretation: The binary-treatment plots ask whether the result depends on the continuous distance scale. The result is strongest and statistically cleaner for larger exposure thresholds, where enforcement deterioration is most plausible.

3.4 Triple-difference outcome regression

To link compensation incentives to product-market outcomes, the paper estimates

$$ProfitMargin_{i,j,t} = \beta_1 Post_t \times \Delta Distance_i \times \Delta PLPS_j \tag{7}$$
$$+ \beta_2 Post_t \times \Delta Distance_i + X_{i,t} + \tau_{j,t} + \gamma_i + \varepsilon_{i,t}, \tag{8}$$

where $\Delta PLPS_j$ is the industry-level reform-induced change in pay sensitivity to local peer performance.

Interpretation: The coefficient β_1 asks whether profitability rises more for exposed firms in industries where CEO pay became more positively tied to local peer performance. A positive estimate links the compensation change to product-market softening rather than merely to contract relabeling.

3.5 Firm-level compensation-change outcome specification

The paper also uses a firm-level indicator

$$D(\Delta PLPS_i > 0), \tag{9}$$

which equals one if the firm significantly increased pay sensitivity to local peer performance after the reform.

Interpretation: This checks that the outcome relation is not only an industry-level pattern. Firms that directly changed their CEO pay loading toward local peers also show stronger product-market outcomes.

3.6 Vesting-horizon regression

For time-vesting equity awards, the paper estimates

$$VestingHorizon_{i,t} = \alpha + \beta\, \Delta Distance_i \times Post_t + X_{i,t} + \gamma_i + \tau_{j,t} + u_{i,t}. \tag{10}$$

Interpretation: Collusion requires patience because a firm can profit in the short run by deviating and undercutting rivals. Longer vesting helps align managers with long-run shareholder gains from stable cooperation.

4 Identification and empirical design

4.1 Why the 2013 DoJ closure is useful

- The closures were driven by federal cost-cutting and office consolidation, not by the compensation policies of individual firms.
- The closure affected the geography of local antitrust enforcement.
- Firms closer to the closed offices and farther from their new covering offices experienced a larger change in enforcement proximity.
- The event affected local product-market monitoring more than national or international antitrust cases.

Interpretation: The research design uses a plausibly exogenous shock to the expected detection probability of collusion. The key comparison is between firms whose antitrust office became farther away and firms whose enforcement office did not materially change.

4.2 Validation of the enforcement shock

- Antitrust case filings dropped in affected states after 2013.
- The drop is especially visible for local antitrust cases, not broad national/international cases.
- Affected states saw the average share of local cases fall from 0.258 to 0.163.
- Unaffected states saw the local-case share rise from 0.241 to 0.303.

Interpretation: These patterns support the idea that the reform reduced local antitrust monitoring. The paper does not merely assume enforcement weakened; it documents that local enforcement activity changed.

4.3 Why local peers matter for identification

- Local peers are more plausible cartel partners than all national industry peers.
- Local peer returns also proxy for common shocks, so the pre-period negative compensation sensitivity validates them as RPE benchmarks.
- The post-reform reversal toward positive local-peer sensitivity is therefore interpretable as a shift from competitive benchmarking toward coordination-friendly incentives.

Interpretation: The design works because the same peer set has two roles: it is a useful benchmark for RPE and a plausible group of product-market collusion partners. A move from negative to more positive loading is economically meaningful.

4.4 Threats to identification and how the paper addresses them

- **Pre-existing state trends.** The authors compare affected and unaffected states and run placebo timing tests.
- **Local economic shocks.** They control for firm characteristics and industry-year fixed effects.
- **Peer-definition sensitivity.** They vary product-peer definitions and locality definitions.
- **Binary versus continuous treatment.** They check many distance thresholds.
- **Treatment-control imbalance.** They use matching and entropy balancing in robustness tests.
- **Geographic clustering.** Standard errors are clustered at the state level in main specifications.
- **Mechanical equity revaluation.** Cash compensation responds more than equity compensation.

Interpretation: The paper cannot randomly assign antitrust office closures, but the empirical design combines shock validation, fixed effects, many robustness checks, and mechanism tests.

4.5 What the paper cannot fully prove

- It does not observe explicit collusion agreements for the main sample.
- It cannot prove boards intended to induce illegal collusion rather than legal softer competition.
- Gross margins are a proxy for market power, not structural markups.
- Stock-return comovement is an indirect measure of coordinated operating behavior.
- Some explicit compensation-contract data are available only for firms covered by Incentive Lab.

Interpretation: The contribution is not direct evidence of cartel formation. It is evidence that compensation contracts move in the direction predicted by a collusion-incentive mechanism after local antitrust enforcement weakens.

5 Tables and figures

5.1 Table 1: Summary statistics

Read:

- The main compensation-pay-performance sample has 10,181 firm-year observations.
- $\Delta Distance$ has mean 1.709 in 100-mile units and median 0, so many firms are unaffected while exposed firms experience sizable distance increases.
- Median total compensation is $\ln(1 + Comp) = 8.303$, median cash compensation is 7.354, and median equity compensation is 7.602.
- The median vesting horizon is 30 months.
- The median firm has 4 local peers and 32 non-local peers.
- Explicit RPE appears in 37.6% of firm-years in the sample for that variable.

Economic message: The sample has enough variation in exposure, local peer networks, and compensation design to test whether antitrust enforcement changes CEO incentives.

Table 1
Summary statistics.

Panel A: Summary statistics

	N	Mean	S.D.	Q1	Median	Q3
ΔDistance	10,181	1.709	3.410	0	0	1.026
Ln(Return)	10,181	0.046	0.471	-0.126	0.102	0.299
Ln(Local peer return)	10,181	0.053	0.389	-0.116	0.098	0.277
Ln(Total compensation)	10,181	8.222	1.035	7.550	8.303	8.957
Ln(Cash compensation)	10,181	7.300	1.088	6.804	7.354	7.909
Ln(Equity compensation)	10,181	6.609	2.941	6.328	7.602	8.464
Vesting horizon	4,584	32.657	10.601	24	30	36
Strategic performance metric (cash)	5,249	0.079	0.270	0	0	0
Strategic performance metric (equity)	5,249	0.009	0.094	0	0	0
Profit margin performance metric (cash)	5,249	0.055	0.228	0	0	0
Profit margin performance metric (equity)	5,249	0.024	0.153	0	0	0
Sales performance metric (cash)	5,249	0.355	0.478	0	0	1
Sales performance metric (equity)	5,249	0.117	0.322	0	0	0
Explicit RPE	7,743	0.376	0.484	0	0	1
Index benchmark	2,917	0.383	0.486	0	0	1
Peer group benchmark	2,917	0.513	0.500	0	1	1
%HP peers in explicit benchmark	1,490	0.176	0.186	0.048	0.107	0.241
%Local peers in explicit benchmark	1,410	0.223	0.295	0	0.100	0.300
Add reciprocal	1,141	0.025	0.157	0	0	0
Gross profit margin	10,172	0.280	4.339	0.252	0.408	0.643
Return comovement	10,180	0.365	0.190	0.233	0.354	0.495
Size	10,179	7.984	1.823	6.653	7.896	9.166
Sales growth	10,176	0.084	0.389	-0.033	0.050	0.147
Ln(Tenure)	10,181	1.805	0.898	1.099	1.946	2.485
Number of local peers	10,181	11.992	17.861	2	4	14
Number of non-local peers	10,181	73.756	97.030	11	32	90
Similarity score of local peers	354,673	0.100	0.069	0.045	0.090	0.139

Panel B: Industry median number of local and non-local peers

FF48	Description	Number of peers	Number of local peers	% Local peers
5	Tobacco Products	8	2	50.00%
15	Rubber and Plastic Products	3	1	50.00%
16	Textiles	2	1	50.00%
23	Automobiles and Trucks	6	2	50.00%
38	Business Supplies	6.5	1.5	41.88%
2	Food Products	4	1	40.00%
21	Machinery	11	4	39.22%
35	Computers	37	10	38.42%
26	Defense	13	4	34.62%
1	Agriculture	3	1	33.33%
	...			
32	Communication	28	3	13.24%
43	Restaurants	25	3	12.25%
45	Insurance	67	7	11.73%
11	Healthcare	84.5	8	11.69%
42	Retail	33	3	11.11%
28	Non-Met. and Indust. Mining	29	1	10.32%
40	Transportation	15	1	10.00%
31	Utilities	61	4	9.83%
44	Banking	332	19	7.02%
29	Coal	119	2	2.78%

Notes: Panel A shows the summary statistics of main variables in the sample. ΔDistance is the increase in geographical distance between a firm's headquarter and

5.2 Tables 2–3: DoJ field-office geography and local antitrust cases

Read:

- Table 2 shows pre-reform coverage of DoJ Antitrust Division field offices. The reform closed Atlanta, Cleveland, Dallas, and Philadelphia.
- The closures affected 23 states and territories.
- Table 3 shows that in affected states the average proportion of local antitrust cases fell from 0.258 in 2008–2012 to 0.163 in 2013–2017, a change of -0.096.
- In unaffected states, the local-case share rose from 0.241 to 0.303, a change of +0.063.
- Convicted local cases fell modestly in both groups, but the larger filed-case pattern supports weakened local enforcement in affected states.

Economic message: These tables validate the treatment. The DoJ closure changed enforcement geography and appears to have reduced local antitrust case activity where offices were removed.

Table 2
States covered by antitrust division field offices.

Field office	States and territories covered by the field offices
Atlanta	Alabama, Florida, Georgia, Mississippi, North Carolina, South Carolina, Tennessee, Puerto Rico, U.S. Virgin Islands
Chicago	Colorado, Illinois, Indiana, Iowa, Kansas, West District of Michigan, Minnesota, Missouri, Nebraska, North Dakota, South Dakota, Wisconsin
Cleveland	Kentucky, Eastern District of Michigan, Ohio, West Virginia
Dallas	Texas, Oklahoma, Louisiana, New Mexico, Arkansas
New York	Connecticut, Maine, Massachusetts, New Hampshire, Northern New Jersey, New York, Rhode Island, Vermont
Philadelphia	Delaware, Maryland, Southern New Jersey, Pennsylvania, Virginia
San Francisco	Alaska, Arizona, California, Hawaii, Idaho, Montana, Nevada, Oregon, Utah, Washington, Wyoming

Notes: This table shows the state coverage of field offices in the U.S. Department of Justice Antitrust Division before the closure of four field

Table 2: field-office coverage

Table 3
Trends in local antitrust cases.

	(1)	(2) Affected states	(3)	(4)	(5) Unaffected states	(6)
	2008-12	2013-17	Diff.	2008-12	2013-17	Diff.
Average proportion of						
1. Local cases	0.258	0.163	-0.096	0.241	0.303	0.063
2. Convicted local cases	0.101	0.080	-0.021	0.209	0.190	-0.019

Table 3: local antitrust case trends

5.3 Figure 1 and Table 4: enforcement validation and RPE benchmark validation

Read:

- Figure 1 shows affected states had a sharp fall in antitrust filings after 2013, while unaffected states did not show the same post-reform drop.
- Table 4 establishes that CEO pay is positively related to own-firm returns and negatively related to peer returns in the pre-benchmarking validation sample.
- In the local-peer sample, local peer return coefficients are negative, around -0.044 to -0.045 in columns (5)–(6).
- This confirms that local product-market peers are economically relevant benchmarks for CEO pay.

Economic message: The paper first verifies both pieces of the design: the reform weakens local enforcement, and peer returns are meaningful compensation benchmarks before studying the post-reform change.

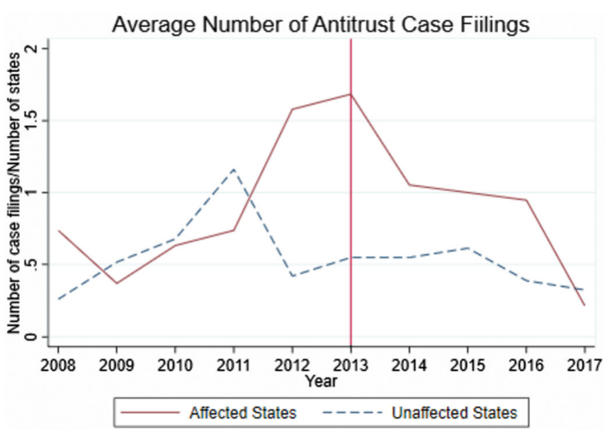


Fig. 1. Trends in antitrust cases. Notes: This figure shows the average number of antitrust case filings per state at the courts of the states that were affected and unaffected by the reform. A state is denoted as "affected" if the firms headquartered in the state experienced an increase in distance from their headquarters to the governing DoJ field offices on average after the reform. Affected states include Alabama, Arkansas, Delaware, Florida, Georgia, Kentucky, Louisiana, Michigan, Mississippi, New Jersey, New Mexico, North Carolina, Ohio, Oklahoma, Pennsylvania, South Carolina, Tennessee, Texas, and West Vir-

Figure 1: antitrust filings in affected versus unaffected states

Table 4
Unconditional sensitivity of pay to performance.

	Ln(Total compensation) (1)	(2)	(3)	(4)	(5)	(6)
Sample:	*full sample*				*with local peer*	
Ln(Return)	0.212***	0.206***	0.202***	0.195***	0.204***	0.193***
	(18.964)	(17.797)	(18.482)	(16.823)	(14.987)	(13.279)
Ln(Peer return)	-0.046***	-0.054***				
	(-3.014)	(-3.076)				
Ln(Local peer return)			-0.035***	-0.031**	-0.044***	-0.045***
			(-2.685)	(-2.251)	(-3.019)	(-2.800)
Local market			0.001	-0.004		
			(0.097)	(-0.273)		
$Size_{t-1}$	0.277***	0.270***	0.273***	0.265***	0.278***	0.267***
	(20.844)	(19.445)	(19.666)	(18.389)	(16.210)	(14.966)
Sales $growth_{t-1}$	0.153***	0.118***	0.157***	0.125***	0.142***	0.113***
	(8.094)	(5.942)	(8.030)	(6.089)	(6.120)	(4.583)
$Ln(Tenure)_t$	0.019**	0.019**	0.018**	0.018**	0.008	0.008
	(2.571)	(2.519)	(2.288)	(2.278)	(0.801)	(0.785)
Constant	5.834***	5.892***	5.860***	5.930***	5.856***	5.945***
	(57.004)	(55.089)	(54.429)	(53.041)	(43.763)	(42.785)
Year FE	YES	NO	YES	NO	YES	NO
Firm FE	YES	YES	YES	YES	YES	YES
SIC2 x Year FE	NO	YES	NO	YES	NO	YES
Adjusted R^2	0.691	0.701	0.688	0.699	0.685	0.697
N	38,526	38,367	36,638	36,491	24,668	24,436

Notes: This table shows the test on unconditional performance sensitivity of total compensation on own and peer returns. The dependent variable is the natural logarithm of one plus total compensation. Ln(Return) refers to the

Table 4: unconditional pay-performance sensitivities

5.4 Table 5: Main effect on CEO total compensation

Read:

- The dependent variable is $\ln(1 + \text{total CEO compensation})$.

- The key peer coefficient is $\Delta Distance \times Post \times \ln(Local\ peer\ return)$.
- In the preferred local-peer specification with SIC2-by-year fixed effects, the coefficient is 0.018 with statistical significance.
- The own-return triple interaction is -0.016, also significant.
- The economic interpretation is that for each additional 100 miles from the antitrust office, CEO pay becomes more positively tied to local peer performance and less tied to own performance after 2013.

Economic message: This is the paper's central empirical result. Weakening local antitrust enforcement changes CEO compensation in a direction that discourages aggressive competition with local peers.

Table 5
DoJ office closures and changes to peer performance sensitivity.

	Ln(Total compensation)					
	(1)	(2)	(3)	(4)	(5)	(6)
Sample:	*full sample*				*with local peers*	
ΔDistance x Post x Ln(Return)	-0.018***	-0.016***	-0.014***	-0.011***	-0.015***	-0.016**
	(-5.075)	(-3.260)	(-4.907)	(-2.705)	(-2.985)	(-2.637)
ΔDistance x Post x Ln(Peer return)	0.022***	0.023***				
	(3.802)	(3.240)				
ΔDistance x Post x Ln(Local peer return)			0.015***	0.014**	0.017***	0.018**
			(3.336)	(2.027)	(3.131)	(2.351)
ΔDistance x Post	-0.001	-0.002	0.013***	0.011***	-0.005	-0.008**
	(-0.289)	(-0.660)	(3.307)	(2.868)	(-1.445)	(-2.064)
Ln(Return)	0.085***	0.088***	0.097***	0.094***	0.100***	0.092***
	(6.371)	(5.859)	(6.840)	(6.279)	(5.571)	(5.429)
ΔDistance x Ln(Return)	0.005*	0.003	0.002	0.001	0.000	0.001
	(1.838)	(0.768)	(0.724)	(0.154)	(0.024)	(0.308)
Post x Ln(Return)	0.166***	0.138***	0.143***	0.124***	0.137***	0.126***
	(7.471)	(5.438)	(5.741)	(4.541)	(3.738)	(3.649)
Ln(Peer return)	0.042	0.003				
	(1.253)	(0.081)				
ΔDistance x Ln(Peer return)	-0.011**	-0.007				
	(-2.313)	(-1.233)				
Post x Ln(Peer return)	-0.117*	-0.103				
	(-1.929)	(-1.501)				
Ln(Local peer return)			-0.008	-0.031	-0.022	-0.053*
			(-0.347)	(-1.327)	(-0.696)	(-1.831)
ΔDistance x Ln(Local peer return)			-0.007	-0.004	-0.006	-0.005
			(-1.515)	(-0.657)	(-1.412)	(-0.804)
Post x Ln(Local peer return)			-0.011	0.002	-0.017	0.012
			(-0.361)	(0.057)	(-0.417)	(0.298)
Local market			-0.054**	-0.048*		
			(-2.188)	(-1.850)		
ΔDistance x Local market			0.017***	0.018***		
			(3.632)	(2.878)		
Post x Local market			0.029	0.019		
			(0.816)	(0.513)		
ΔDistance x Post x Local market			-0.017***	-0.017**		
			(-3.007)	(-2.373)		
$Size_{t-1}$	0.271***	0.270***	0.271***	0.271***	0.297***	0.290***
	(12.680)	(11.973)	(13.107)	(12.380)	(11.576)	(9.578)
$Sales\ growth_{t-1}$	0.028*	0.017	0.026*	0.016	0.013	0.006
	(1.729)	(1.121)	(1.698)	(1.107)	(0.687)	(0.331)
Ln(Tenure)	0.045***	0.043***	0.044***	0.043***	0.037***	0.033**
	(3.956)	(3.750)	(3.921)	(3.756)	(2.847)	(2.556)
Constant	5.999***	6.012***	5.998***	6.008***	5.797***	5.862***
	(36.181)	(35.053)	(36.527)	(35.713)	(29.094)	(25.018)
Year FE	YES	NO	YES	NO	YES	NO
Firm FE	YES	YES	YES	YES	YES	YES
SIC2 x Year FE	NO	YES	NO	YES	NO	YES
Adjusted R^2	0.755	0.757	0.755	0.757	0.760	0.761
N	13,946	13,894	13,946	13,894	10,181	10,109

Notes: This table reports the regression results on changes in pay-performance sensitivity in response to the DoJ field office closures. The dependent variable is the natural logarithm of one plus total compensation. Post is a dummy variable that takes the value of one if the year is on or after 2013, and zero otherwise. ΔDistance is the increase in geographical distance between a firm's headquarter and its governing antitrust office after the closure of four field offices (Atlanta, Cleveland, Dallas, and Philadelphia) in 100 miles. Ln(Return) refers to the natural logarithm of one

Table 6
Compensation components.

	(1)	(2)	(3)	(4)
	Ln(Cash compensation)		Ln(Equity compensation)	
Sample:	*with local peers*			
ΔDistance x Post x Ln(Return)	-0.013*	-0.014**	-0.006	0.000
	(-1.981)	(-2.273)	(-0.199)	(0.013)
ΔDistance x Post x Ln(Local peer return)	0.011**	0.014***	0.008	0.019
	(2.556)	(4.434)	(0.427)	(0.547)
ΔDistance x Post	-0.005	-0.012***	-0.022	-0.031*
	(-1.443)	(-2.688)	(-1.466)	(-1.913)
Ln(Return)	0.186***	0.181***	0.112	0.062
	(5.680)	(6.038)	(1.609)	(0.977)
ΔDistance x Ln(Return)	-0.001	-0.002	-0.042*	-0.037*
	(-0.115)	(-0.669)	(-1.846)	(-1.713)
Post x Ln(Return)	0.174***	0.170***	0.224**	0.260***
	(4.323)	(4.254)	(2.145)	(2.978)
Ln(Local peer return)	0.048*	0.023	-0.205	-0.291**
	(1.749)	(0.734)	(-1.570)	(-2.665)
ΔDistance x Ln(Local peer return)	-0.008**	-0.006	0.018	0.024
	(-2.583)	(-1.676)	(0.788)	(0.809)
Post x Ln(Local peer return)	0.026	0.048	0.027	0.056
	(0.894)	(1.341)	(0.131)	(0.254)
$Size_{t-1}$	0.154***	0.161***	0.574***	0.547***
	(3.299)	(3.946)	(5.564)	(4.665)
$Sales\ growth_{t-1}$	0.030	0.024	0.049	0.051
	(1.634)	(1.335)	(0.758)	(0.760)
Ln(Tenure)	0.060***	0.057***	-0.105*	-0.114**
	(6.416)	(4.789)	(-1.985)	(-2.223)
Constant	5.953***	5.911***	2.261***	2.498***
	(15.972)	(18.042)	(2.785)	(2.717)
Year FE	YES	NO	YES	NO
Firm FE	YES	YES	YES	YES
SIC2 x Year FE	NO	YES	NO	YES
Adjusted R^2	0.681	0.690	0.511	0.510
N	10,183	10,111	10,181	10,109

5.5 Table 6: Cash versus equity compensation

Read:

- The positive local-peer sensitivity effect is concentrated in cash compensation.
- In the cash-compensation specification with industry-year fixed effects, $\Delta Distance \times Post \times \ln(Local\ peer\ return)$ is 0.014 and highly significant.
- In the equity-compensation specifications, the corresponding coefficients are not statistically significant.
- The own-return interaction is negative for cash compensation and weak or insignificant for equity compensation.

Economic message: Boards appear to actively adjust flexible pay components rather than simply benefiting from mechanical equity revaluation as collusive rents raise stock prices.

5.6 Figure 2 and Table 7: Robustness

Read:

- Figure 2 replaces the continuous distance variable with a series of binary treatment cutoffs.
- The local-peer coefficient is generally positive and the own-return coefficient is generally negative, especially for higher exposure thresholds.
- Table 7 summarizes robustness checks: distance cutoffs, matching, entropy balancing, placebo reforms, alternative peer definitions, alternative locality definitions, added fixed effects, Jackknife clustered standard errors, and winsorization choices.
- The main sign pattern survives these checks.

Economic message: The baseline result is not driven by one arbitrary treatment measure, one peer definition, one matching choice, or one clustering procedure.

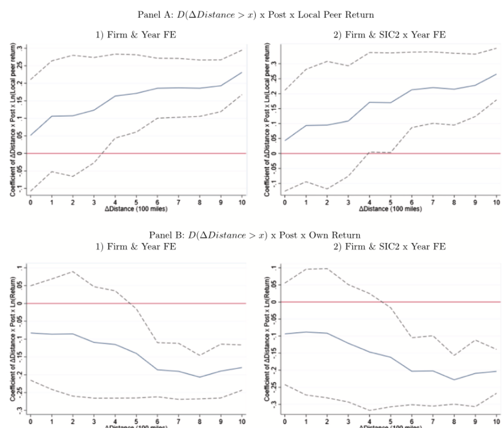


Fig. 2. Binary Treatment. Notes: These figures show the coefficient estimates from the robustness of baseline regression using binary treatment variables. The regression specification follows columns (5)-(6) of Table 5 with $\Delta Distance$ replaced by a dummy indicator that $\Delta Distance$ is above a threshold x as in the horizontal axis. x is within the range of 0 to 1000 miles. Panel A plots the coefficient on the triple interaction of the logged local peer return, post-reform dummy, and the binary treatment. Panel B plots the coefficient on the triple interaction of the logged own firm return, post-reform dummy, and the binary treatment. In each panel, we present findings using two different sets of fixed effects: 1) firm and year FE, and 2) firm and joint fixed effect between year and SIC 2-digit industry. The solid lines plot the point estimation of coefficients, and the dashed lines plot the 95% confidence intervals. All variables are winsorized at 0.5% and 99.5% levels. The sample spans from 2008 to 2017 and contains Execucomp firms with local peers.

Table 7
Alternative specifications and robustness tests.

	(1)	(2)
	ΔDistance x Post x Ln(Local peer return)	
	Year FE	SIC2 x Year FE
Panel A: Binary treatment variable		
1. D(ΔDistance > 400))	0.164***	0.171**
	(2.678)	(2.019)
Panel B: Propensity score matching		
2. ΔDistance> 400 vs. ≤ 400	0.027***	0.028*
	(2.978)	(1.737)
3. ΔDistance> 0 vs. = 0	0.037***	0.041***
	(3.729)	(3.009)
Panel C: Entropy balancing		
4. ΔDistance> 400 vs. ≤ 400	0.021***	0.027***
	(4.517)	(3.196)
5. ΔDistance> 0 vs. = 0	0.021***	0.024***
	(3.993)	(2.715)
Panel D: Placebo test		
6. Pseudo shock in 2006	-0.008	-0.006
	(-0.748)	(-0.475)
Panel E: Alternative definition of peers - Hoberg-Phillips similarity score		
7. Within top 30%	0.015**	0.014
	(2.228)	(1.444)
8. Within top 60%	0.019***	0.021**
	(2.876)	(2.341)
9. Within top 100%	0.011	0.011
	(1.621)	(0.907)
Panel F: Alternative definitions of peers - Other classifications		
10. Similar size and book-to-market	0.015**	0.012
	(2.141)	(1.039)
11. Cross-price demand elasticity	0.021**	0.023*
	(2.578)	(1.838)
12. Factset Revere peers	0.065***	0.039*
	(3.978)	(1.907)
13. Analyst-based peer group	0.088*	0.225***
	(1.771)	(2.735)
Panel G: Alternative definition of locality		
14. Local peers within 100 miles	0.023***	0.025***
	(4.809)	(3.375)
15. Local peers within 300 miles	0.016**	0.016*
	(2.443)	(1.897)
16. Local peers within 400 miles	0.019***	0.018**
	(2.979)	(2.548)
Panel H: Control for non-local peers		
17. ΔDistance x Post x Ln(Local peer return)	0.020***	0.026**
	(3.006)	(2.452)
ΔDistance x Post x Ln(Non-local peer return)	-0.003	-0.014
	(-0.501)	(-1.353)
Panel I: Other regression choices		
18. Without firms who got closer	0.015**	0.017**
	(2.684)	(2.086)
19. Winsorize at 1% and 99% level	0.017***	0.018**
	(2.966)	(2.213)
20. Poisson regression	0.023***	0.018*
	(3.340)	(1.752)

Table 7 (*continued*)
Notes: This table shows ten different sets of robustness tests using alternative specifications to estimate the coefficient of β_2 reported in Table 5, columns (5)-(6). In Panel A, row 1, we replace ΔDistance with a binary variable that equals one if the change in distance to the field office is more than or equal to 400 miles, and zero otherwise. In Panel B, we match the treated and control firms with the closest propensity of being treated based on logistic regression of the treated dummy on own and local peer returns, firm size, sales growth rate, CEO tenure, and industry dummies using the 2012 sample. In row 2, the treated firms are those with $\Delta Distance$ above 400 miles; in row 3, the treated firms are those with $\Delta Distance$ above 0 miles. In Panel C, rows 4-5, we apply the entropy balancing method to reweigh these firm characteristics when estimating the treatment effect. In Panel D, row 6, we conduct the placebo test by taking 2006 as the pseudo event year. In Panel E, rows 7-9, local peers are defined as the firms within the top 30, 60, and 100% of the Hoberg-Phillips similarity scores. In Panel F, we use alternative classifications to define local peers. In row 10, we use size and book-to-market screening, choosing the closest half of local peers based on the closeness of the Mahalanobis distance using the market capitalization and book-to-market. In row 11, we use cross-price demand elasticities with focal firms (Pellegrino (2023)), defining local peers are the ones within the top tertile of cross-price demand elasticities with focal firms each year. In row 12, we use the Factset Revere classification. In row 13, we use an analysts-based peer group (Kaustia and Rantala (2021)), where local peer return is defined as the average stock performance of local peers who are followed by common analysts weighted on the number of analysts each year. In Panel G, rows 14-16, we define local peers as the firms headquartered within 100, 300, and 400 miles of the focal firm. In Panel H, row 17, we additionally control for average non-local peer return and their interaction terms. In Panel I, row 18, we drop the sample of firms who got closer to the governing DoJ offices after 2013, in row 19, all variables are winsorized at the 1% and 99% level, in row 20, we estimate Poisson regression. In Panel J, rows 21-24, standard errors are clustered at the levels of firm, ZIP code, SIC 2-digit, and pre-shock covering DoJ office region levels. The dependent variables are the natural logarithm of one plus total compensation. Post is a dummy variable that equals one if the year is on or after 2013, and zero otherwise. ΔDistance is the increase in geographical distance between a firm's headquarter and its governing antitrust office after the closure of four field offices (Atlanta, Cleveland, Dallas, and Philadelphia) in 100 miles. Ln(Local peer return) is the natural logarithm of one plus average annual stock return of local product market peers that are the Hoberg-Phillips peers with similarity score within the top 70% and headquartered within 200 miles of the focal firm. SIC2 x Year FE is the joint fixed effect between year and SIC 2-digit industry. The data includes firms with local peers in Execucomp and spans from 2008 to 2017 (except for the sample for Panel D spans from 2001 to 2010). All the variables are winsorized at the 0.5% and 99.5% levels (except for row 19). Except for Panel J, robust t-statistics based on standard errors clustered at the state level are reported in parentheses. The full tables are reported in Internet Appendix Table IA2, Table IA3, Table IA4, Table IA5, Table IA6, Table IA7, Table IA8, Table IA9, and Table IA10.

substitute goods estimated using the cross-price demand elasticity in Pellegrino (2023); (b) we use peer firms as recorded in Factset Supply Chain Relationships (formerly, Revere); (c) we define peers as the firms covered by common analysts (Kaustia and Rantala, 2021). The findings are reported in Panel F of Table 7 and in Panel B of Internet Appendix Table IA7.

Finally, we investigate different definitions of locality in Table 7, Panel G, and in Internet Appendix Table IA8. In our baseline tests, we define local firms as the ones headquartered within 200 miles. Alternatively, we define local peers to be the firms headquartered within 100, 300, and 400 miles, and find an increase in pay-to-peer-performance

5.7 Table 8: Heterogeneity across competition, governance, geography, and CEO incentives

Read:

- The local-peer sensitivity effect is much stronger in industries with strategic complements: 0.040 versus 0.007 for strategic substitutes.
- It is stronger in industries where the largest eight firms account for a high revenue share.
- It is stronger where a larger share of industry firms are public, making peer stock returns more informative.
- It is stronger for firms with geographically concentrated sales or concentrated state mentions in 10-Ks.
- It is stronger when boards are less co-opted, consistent with shareholder-oriented boards adjusting pay.
- It is stronger for older CEOs and for firms in states without Inevitable Disclosure Doctrine, consistent with larger wedges between managerial and shareholder incentives.

Economic message: The cross-sectional evidence lines up with the collusion story: the effect is strongest where collusion is easier to coordinate, easier to infer from peer returns, and more dependent on aligning managerial incentives.

Table 8
Heterogeneity.

	(1)	(2)	(3)
	ΔDistance x Post x Ln(Local peer return)		
1. Competition mode	Strategic substitutes	Strategic complements	Difference
	0.007	0.040***	-0.033***
	(0.731)	(3.339)	(3.289)
2. Revenue of largest 8 firms in the NAICS	Low	High	Difference
	0.016	0.046***	-0.030
	(0.914)	(2.907)	(-1.231)
3. Fraction of public firms	Low	High	Difference
	0.020	0.038***	-0.017*
	(1.646)	(4.297)	(-1.713)
4. Concentration of sales across the states	Low	High	Difference
	0.010	0.081**	-0.070**
	(0.652)	(2.240)	(-2.525)
5. Concentration of states mentioned in 10Ks	Low	High	Diff
	0.002	0.043***	-0.041***
	(0.166)	(3.234)	(-2.962)
6. Co-opted board	High	Low	Difference
	-0.004	0.038***	-0.042***
	(-0.490)	(4.370)	(-4.083)
7. CEO age	Young	Old	Difference
	0.011	0.032***	-0.021***
	(1.493)	(3.251)	(-2.774)
8. Inevitable Disclosure Doctrine	IDD	Non-IDD	Difference
	-0.030*	0.028***	-0.058***
	(-1.941)	(5.086)	(-3.597)

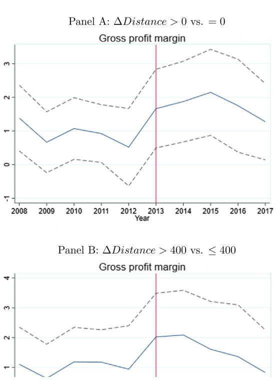
Notes: This table presents ten different heterogeneity tests. In row 1, we partition firms into those who operate in an industry in which firms compete as strategic complements or strategic substitutes following Kedia (2006). In row 2, we split the sample based on the revenue percentage of the largest 8 firms over all firms for each NAICS 4-digit industry in 2012. Firms in the "High" ("Low)" group operate in NAICS industry where the percentage of revenue

5.8 Figure 3 and Table 9: Product-market outcomes

Read:

- Figure 3 shows treated firms' gross profit margins rose relative to control firms after 2013.
- This pattern appears under both treatment definitions: any increase in distance and an increase above 400 miles.
- Table 9 links compensation changes to outcomes.
- The interaction $Post \times \Delta Distance \times \Delta PLPS_{sic2}$ is positive for gross profit margins and return comovement.
- The firm-level measure $D(\Delta PLPS_{firm} > 0)$ also produces positive triple interactions, especially for return comovement.

Economic message: Contract changes are associated with real product-market outcomes. Firms that become more exposed and more coordination-friendly in CEO pay also look more profitable and more correlated with local peers.



in the industries in which the treated firms were more responsive to the reform in terms of changing their CEO pay. We thus construct an industry-level measure of the reform-induced change in sensitivity of CEO pay to local peer performance. Specifically, we estimate the β_{2j} coefficient from separate regressions (1) for each SIC 2-digit industry, j, and refer to it as change in pay-to-local-peer-performance sensitivity, namely $\Delta PLPS_j$. We then estimate a regression with the following specification using the sample of firms with local peers:

$$\begin{aligned} Profit\ Margin_{i,j,t} = & \beta_1 \times Post_t \times \Delta Distance_i \times \Delta PLPS_j \\ & + \beta_2 \times Post_t \times \Delta Distance_i + X_{i,t} + \tau_{j,t} + \gamma_i + \epsilon_{i,t} \end{aligned} \quad (2)$$

where for each firm i of industry j and year t, $Profit\ Margin_{i,j,t}$ corresponds to the gross profit margin, i.e., the gross profit over revenue. $Post_t$ is the dummy indicator for the years on and after 2013. $\Delta Distance_i$ refers to the increase in geographical distance between firm's headquarter and a governing antitrust field office after the reform in 100-mile units. We control for firm characteristics, firm fixed effects, and the joint year and industry fixed effects at the SIC 2-digit level.

This triple difference methodology allows us to control for the general industry-wide trends as we rely on the incremental post-reform change in the peer performance sensitivity by the treated group as compared to the same-industry control group. Thus, the estimate of β_1 in specification (2) captures the heterogeneity in reform-induced change in gross profit margins across industries associated with different degrees of CEO compensation adjustments. Since a more positive $\Delta PLPS_j$ reflects a greater anti-competitive incentive provision, we expect β_1 to be positive. We find supporting findings in column (1) of Table 9: the increase in profitability induced by the reform was concen-

Figure 3: gross profit margins

Table 9
Incentive alignment and firm outcomes.

	(1)	(2)	(3)	(4)
	Gross profit margin		Return comovement	
Post x ΔDistance	-0.002*	-0.002*	0.001	0.000
	(-1.853)	(-1.856)	(1.296)	(0.461)
Post x ΔDistance x $\Delta PLPS_{sic2}$	0.001***		0.001***	
	(5.937)		(2.706)	
Post x $D(\Delta PLPS_{firm} > 0)$		0.000		-0.023
		(0.006)		(-1.524)
Post x ΔDistance x $D(\Delta PLPS_{firm} > 0)$		0.002*		0.005***
		(1.937)		(3.248)
$Size_{t-1}$	-0.005	-0.006	0.020***	0.020***
	(-1.131)	(-1.156)	(4.428)	(4.508)
$Sales\ growth_{t-1}$	0.007	0.009	0.003	0.002
	(1.309)	(1.296)	(0.717)	(0.487)
Ln(Tenure)	0.002	0.002	0.003	0.004
	(1.273)	(0.854)	(1.075)	(1.320)
Constant	0.509***	0.517***	0.193***	0.193***
	(13.393)	(12.592)	(5.322)	(5.123)
Firm FE	YES	YES	YES	YES
SIC2 x Year FE	YES	YES	YES	YES
Adjusted R^2	0.931	0.930	0.664	0.668
N	7,985	7,195	8,167	7,362

Notes: This table demonstrates the variation in firm outcome resulting from the DoJ reform as firms adjust their pay-performance sensitivity in response to the reform. The dependent variables in the regressions are the gross profit margin and the return comovement. The gross margin is defined as the gross profit divided by sales. The return comovement is the average annual correlation of weekly stock market returns between the firm and its local peers. The local peer firms are defined as the firms with Hoberg-Phillips product similarity score within the top 70% and are headquartered within 200 miles of the focal firm. Post is a dummy indicator that takes the value of one for the years on and after 2013, and zero otherwise. ΔDistance is the increase in geographical distance between a firm's headquarter and its governing antitrust office after the closure of four field offices (Atlanta, Cleveland, Dallas, and

Table 9: incentive alignment and firm outcomes

5.9 Table 10: Explicit RPE peer groups

Read:

- The paper studies whether firms change explicit relative performance evaluation provisions, not only implicit pay-performance sensitivity.
- Explicit RPE adoption does not change strongly in response to the reform.
- Conditional on explicit RPE, exposed firms reduce the proportion of Hoberg-Phillips product-market peers in benchmark groups.
- The reduction is especially relevant for local peers and reciprocal peer-group overlap, though some estimates are less precise because the Incentive Lab sample is smaller.

Economic message: Boards do not simply add or remove RPE wholesale. Instead, when explicit RPE exists, they adjust the composition of peer groups in a less competition-intensifying direction.

Table 10
Explicit relative performance evaluation.

Panel A: Explicit RPE and benchmark choice

	(1)	(2)	(3)	(4)	(5)	(6)
	Explicit RPE		Index benchmark		Peer group benchmark	
Sample:	*with local peers*		*with local peer & explicit RPE*			
ΔDistance x Post	0.002	0.001	0.003	0.002	-0.003	-0.003
	(0.993)	(0.341)	(0.775)	(0.348)	(-0.619)	(-0.273)
Size_{t-1}	0.006	-0.008	0.100***	0.059	-0.034	-0.020
	(0.211)	(-0.352)	(3.013)	(1.405)	(-0.849)	(-0.490)
$\text{Sales growth}_{t-1}$	-0.021*	-0.024**	-0.005	0.021	-0.046	-0.049
	(-1.614)	(-1.938)	(-0.138)	(0.383)	(-0.931)	(-0.792)
Ln(Tenure)_t	-0.014	-0.017*	-0.020*	-0.022*	0.014	0.021
	(-1.452)	(-1.792)	(-1.849)	(-1.878)	(1.407)	(1.503)
Constant	0.407**	0.535**	-0.520*	-0.139	0.827**	0.687*
	(2.302)	(2.225)	(-1.697)	(-0.349)	(2.185)	(1.734)
Year FE	YES	NO	YES	NO	YES	NO
Firm FE	YES	YES	YES	YES	YES	YES
SIC2 x Year FE	NO	YES	NO	YES	NO	YES
Adjusted R^2	0.608	0.605	0.682	0.687	0.638	0.640
N	6,050	5,949	2,479	2,349	2,479	2,349

Panel B: Peer group composition

	(1)	(2)	(3)	(4)	(5)	(6)
	%HP peers		%Local peers		Add reciprocal	
Sample:	*with local peer & explicit peer group benchmark*					
ΔDistance x Post	-0.003**	-0.004**	-0.005**	0.001	-0.001	-0.006**
	(-2.208)	(-2.542)	(-2.218)	(0.189)	(-0.494)	(-2.109)
Size_{t-1}	-0.001	-0.009	-0.010	-0.024	-0.022	-0.043
	(-0.041)	(-0.553)	(-0.424)	(-0.821)	(-1.188)	(-1.420)
$\text{Sales growth}_{t-1}$	-0.003	0.018	-0.007	0.006	-0.008	-0.001
	(-0.285)	(1.337)	(-0.273)	(0.281)	(-0.397)	(-0.065)
Ln(Tenure)_t	-0.003	-0.002	0.021*	0.011	0.008	0.009*
	(-0.520)	(-0.338)	(2.009)	(0.947)	(1.503)	(1.814)
Constant	0.200	0.274	0.296	0.420	0.222	0.443
	(0.991)	(1.651)	(1.305)	(1.550)	(1.262)	(1.494)
Firm FE	YES	YES	YES	YES	YES	YES
Year FE	YES	NO	YES	NO	YES	NO
SIC2 x Year FE	NO	YES	NO	YES	NO	YES
Adjusted R^2	0.753	0.783	0.670	0.689	-0.051	-0.206
N	1,266	1,123	1,206	1,075	967	849

Notes: This table shows the regression results on the likelihood of granting CEOs incentive plans with explicit relative performance evaluation (RPE) provisions, the choice of using an index or specifying a peer group as the relative performance benchmark, and the composition of the specified peer group. The dependent variables of Panel A include the dummy indicator of having explicit RPE provision and the dummy indicators of using an index or specified peer group as the performance benchmark (conditional on having explicit RPE plans). In Panel B, the first dependent variable is the proportion of product market peers included as the explicit peer group benchmark, where product market peers refer to the Hoberg-Phillips peers with product similarity score within the top 70%; the second dependent variable is the proportion of local peers included as the explicit peer group benchmark, where local peers refer to the Hoberg-Phillips peers with product similarity score within the top 70% and headquartered with 200 miles; the third dependent variable is a dummy indicator of a net increase of reciprocal peers, i.e., the number of additional reciprocal peers is larger than the number of additional non-reciprocal peers, in the explicit peer group benchmark. ΔDistance is the increase in geographical distance between a firm's headquarter and its governing antitrust office after the closure of four field offices (Atlanta, Cleveland, Dallas, and Philadelphia) in 100 miles. Size is the natural logarithm of one plus total assets. Sales growth is the ratio of current year sales minus previous year sales and previous year sales. Ln(Tenure) is the natural logarithm of one plus the years since the executive assumes their CEO position. SIC2 x Year FE is the joint fixed effect between year and SIC 2-digit industry. All the variables are winsorized at the 0.5% and 99.5% levels. The sample for Panel A, columns (1)-(2), includes firms with local peers and are covered by Incentive Lab, the sample for Panel A, columns (3)-(6), further requires the firms to have explicit RPE grants, the sample for Panel B, columns (1)-(4), further requires the

5.10 Tables 11–12: Performance metrics and vesting horizons

Read:

- Table 11 shows affected firms increase use of strategic metrics in cash plans and profit-margin metrics in cash plans.
- The use of sales metrics in stock plans declines significantly.
- The pattern suggests less emphasis on expanding output and more emphasis on margins or discretionary strategic goals.
- Table 12 shows exposed firms lengthen CEO equity vesting horizons.
- The coefficient on $Post \times \Delta Distance$ is 0.178 months per 100 miles in the continuous treatment specification.
- The binary treatment $D(\Delta Distance > 400)$ produces a 2.306-month increase in vesting horizon.

Economic message: Compensation changes are broader than pay-to-peer-return loadings. Affected firms move toward long-horizon, margin-oriented, and strategic incentives that can sustain less aggressive product-market behavior.

Table 11
Performance metrics in incentive plans.

	(1)	(2)	(3)	(4)	(5)	(6)
	Strategic		Profit margin		Sales	
	Cash	Stock	Cash	Stock	Cash	Stock
ΔDistance x Post	0.004*	-0.000	0.002**	-0.000	0.000	-0.006***
	(1.865)	(-0.087)	(2.137)	(-0.350)	(0.055)	(-3.714)
Size_{t-1}	-0.022**	-0.001	-0.016*	-0.012	-0.026	0.033*
	(-2.122)	(-0.243)	(-1.693)	(-1.010)	(-1.623)	(1.926)
Sales growth_{t-1}	0.006	0.008**	0.004	0.002	0.011	-0.016*
	(0.307)	(2.340)	(1.077)	(0.682)	(0.508)	(-1.820)
Ln(Tenure)	0.003	0.001	-0.005	-0.004	-0.002	0.006
	(0.515)	(0.283)	(-1.269)	(-1.484)	(-0.227)	(0.851)
Constant	0.263***	0.019	0.207**	0.139	0.583***	-0.177
	(2.892)	(0.384)	(2.413)	(1.272)	(4.111)	(-1.191)
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Adjusted R^2	0.390	0.306	0.559	0.339	0.725	0.501
N	5,442	5,429	5,442	5,429	5,442	5,429

Notes: This table presents the changes in the use of performance metrics in CEO incentive plans after the relocation of DoJ field offices. The three dependent variables are the dummy indicators of whether a firm mentions the following terms as the performance metrics of its CEO's cash or stock incentive plans: "strateg*" for columns (1)-(2), "profit margin" for columns (3)-(4), and "sales", "gross revenue", "same store sales", or "revenue related" for column (5)-(6). Post is a dummy variable indicating whether the year is on or after 2013. ΔDistance is the increase in geographical distance between a firm's headquarter and its governing antitrust office after the closure of four field offices (Atlanta, Cleveland, Dallas, and Philadelphia) in 100 miles. Size is natural logarithm of one plus total assets. Sales growth is the ratio of current year sales minus previous year sales and previous year sales. Ln(Tenure) is the natural logarithm of one plus the years since the executive assumes their CEO position. All the variables are winsorized at the 0.5% and 99.5% levels. The data includes firms with local peers that are covered by Incentive Lab database and spans from 2008 to 2017. Robust t-statistics based on standard errors clustered at the state level are reported in parentheses.

in the states in which courts did not recognize IDD, i.e. where the CEOs were facing a more mobile labor market and potentially cared more for their shorter-term reputations relative to long-term shareholder value. These findings suggest that the reform induced the affected firms to provide long-term incentives that could potentially foster collusion, es-

notion that both the provision of incentives and the overall level of pay decreased once aggressive competition was no longer preferred.

Last, in Panel B of Internet Appendix Table IA14, we study the composition of CEOs' compensation packages, as divided into four parts: salary, other cash incentive plans, equity incentive plans, and other

Table 11: performance metrics

Table 12
Vesting horizon.

	Vesting horizon (1)	(2)	(3)	(4)	(5)
Sample:	*full sample*			*without IDD*	*with IDD*
Post x ΔDistance	0.178**		0.178**	0.172***	0.287
	(2.496)		(1.997)	(3.808)	(0.923)
Post x D(ΔDistance> 400)		2.306***			
		(2.798)			
Size_{t-1}	-0.463	-0.471	-0.463	-1.070	-0.209
	(-0.981)	(-1.005)	(-0.807)	(-1.733)	(-0.266)
Sales growth_{t-1}	0.163	0.155	0.163	0.471	-0.058
	(0.285)	(0.271)	(0.322)	(0.509)	(-0.229)
Ln(Tenure)	-0.988***	-0.986***	-0.988***	-0.995***	-1.346**
	(-4.399)	(-4.423)	(-3.905)	(-3.520)	(-2.468)
Constant	38.157***	38.190***	38.157***	43.594***	36.278***
	(9.042)	(9.144)	(7.482)	(8.215)	(5.095)
Firm FE	YES	YES	YES	YES	YES
SIC2 x Year FE	YES	YES	YES	YES	YES
Cluster	State	State	Firm	State	State
Adjusted R^2	0.560	0.561	0.560	0.540	0.599
N	4,560	4,560	4,560	2,992	1,392

Notes: This table presents the effect of DoJ field office relocation on the vesting horizon of CEO's time-vesting equity plans. The dependent variable is determined as follows: For each incentive plan, we calculate the duration in months until the last vesting date for cliff vesting plans, and we compute the average duration in months between the first and last vesting dates for ratable vesting plans. In cases where multiple plans are granted in a single year, the vesting horizon is defined as the maximum number of months across all plans. Post is a dummy variable that equals one if the year is on or after 2013, and zero otherwise. ΔDistance is the increase in geographical distance between a firm's headquarter and its governing antitrust office after the closure of four field offices (Atlanta, Cleveland, Dallas, and Philadelphia) in 100 miles. D(ΔDistance> 400) is a dummy variable that takes the value of one if a firm experienced more than 400-mile increase in geographic distance. "With IDD" refers to the sample of firms who were located in

Table 12: vesting horizon

5.11 Table 13: Evidence from convicted cartels

Read:

- The paper separately studies public firms that participated in convicted cartels, using Connor's cartel data.
- Before cartel initiation, CEO pay is negatively related to cartel-peer returns, similar to conventional RPE.
- After cartel initiation, pay becomes more positively related to cartel-peer performance; $Post \times \ln(Cartel\ peer\ return)$ is positive and statistically significant.
- Compensation benchmark groups overlap substantially with cartel peers at the general compensation level, but less in formal RPE groups.
- During active cartel years, 20.67% of cases include at least one cartel peer in the general compensation benchmark, compared with 10.10% in formal RPE benchmarks.

Economic message: The convicted-cartel evidence supports the main interpretation: when collusion is actually observed, CEO compensation becomes less anti-peer and more peer-aligned.

Table 13

Convicted cartel peers and compensation schemes.

Panel A: Sensitivity of CEO compensation to cartel peer performance

	(1)	(2)	(3)	(4)	(5)	(6)
			Ln(Total compensation)			
Ln(Return)	0.356***	0.339***	0.694***	0.669***	0.152	0.137
	(3.238)	(3.046)	(5.170)	(5.027)	(0.827)	(0.702)
Ln(Cartel peer return)	-0.059***	-0.057***	-0.158***	-0.156***	-0.111***	-0.111***
	(-2.775)	(-2.703)	(-2.864)	(-2.836)	(-2.927)	(-2.864)
Post x Ln(Return)			-0.444**	-0.425**	-0.405*	-0.393
			(-2.233)	(-2.196)	(-1.759)	(-1.620)
Post x Ln(Cartel peer return)			0.136**	0.138**	0.093**	0.096**
			(2.137)	(2.256)	(2.115)	(2.269)
Post x Ln(Non-cartel peer return)					-0.077	-0.075
					(-1.392)	(-1.432)
Post			-0.001	0.215	-0.308	-0.015
			(-0.006)	(1.513)	(-1.513)	(-0.129)
Ln(Non-cartel peer return)					0.096**	0.102**
					(2.324)	(2.422)
$Size_{t-1}$	-0.331***	-0.330***	-0.312***	-0.314***	-0.324***	-0.320***
	(-3.834)	(-3.674)	(-3.662)	(-3.508)	(-3.957)	(-3.632)
Sales $growth_{t-1}$	0.087	0.089	0.088*	0.092	0.050**	0.054**
	(1.613)	(1.562)	(1.683)	(1.627)	(2.192)	(2.226)
$Ln(Tenure)_t$	0.073*	0.074*	0.076*	0.075*	0.125***	0.122***
	(1.839)	(1.810)	(1.973)	(1.867)	(3.004)	(2.702)
Constant	12.141***	12.142***	11.935***	11.816***	12.535***	12.300***
	(13.365)	(12.799)	(13.049)	(12.379)	(14.099)	(12.996)
Year FE	YES	YES	YES	YES	YES	YES
Firm x Cartel FE	YES	NO	YES	NO	YES	NO
Firm FE	NO	YES	NO	YES	NO	YES
Adjusted R^2	0.472	0.500	0.478	0.504	0.516	0.547
N	1,642	1,643	1,642	1,643	1,264	1,269

Panel B: Overlap of cartel peers with compensation peers

	(1)	(2) At least one peer	(3) Fraction peers	(4) N
	Overlap at the time of cartel			
General compensation benchmark	86	20.67%	7.81%	416
Relative performance evaluation benchmark	42	10.10%	3.67%	416
Difference		10.58%***	4.14%***	
		(5.803)	(4.838)	
	Overlap over the entire sample period			
General compensation benchmark	252	50.70%	20.35%	497
Relative performance evaluation benchmark	117	23.54%	8.09%	497
Difference		27.16%***	12.26%***	
		(12.095)	(10.317)	

Notes: Panel A shows the sensitivity of CEO compensation to the performance of own firm and the other cartel members in the convicted cartels, as collected from Connor (2014). The sample contains public firms with a history

6 Section-by-section detailed summary

6.1 Introduction

- The paper begins with a corporate-governance puzzle: shareholders may want managers to pursue strategies that raise firm value, but some profitable strategies can reduce consumer welfare.
- Collusion is a clear example. A cartel can raise profits, but managers face personal criminal liability, imprisonment risk, reputational costs, and career concerns.
- Because shareholders and managers face different costs and horizons, compensation can be used to align managers with shareholder preferences for softer competition.
- The authors argue that a decline in antitrust enforcement should increase shareholders' demand for collusion-friendly compensation.
- The 2013 DoJ field-office closures provide the empirical setting.
- The prediction is not merely that CEO pay rises. The prediction is about the *structure* of pay: peer-performance sensitivity should become more positive, own-performance sensitivity should weaken, and incentive horizons should lengthen.

6.2 Data section

- The authors use ExecuComp to measure CEO compensation and Incentive Lab to observe explicit performance benchmarks and vesting terms.
- Hoberg-Phillips product similarity links firms to product-market rivals.
- Local peers combine product similarity with geographic proximity.
- This local-peer definition is important because the reform is about local enforcement and because many collusive arrangements are local in nature.
- The sample is restricted to firms with available compensation, stock return, financial, and peer information.
- Summary statistics show meaningful cross-sectional variation in exposure and local peer networks.

6.3 Identification section

- The four office closures were part of a cost-saving and consolidation initiative.
- The authors validate the shock by showing that affected states experienced a drop in local antitrust filings after 2013.
- The paper argues that the reform changed the expected detection probability for local product-market collusion.
- The empirical design compares firms before and after 2013, with firm fixed effects and year or industry-year fixed effects.
- The continuous treatment allows more affected firms to have larger treatment intensity.
- Robustness tests address concerns about state trends, treatment definition, peer construction, and sample imbalance.

6.4 Main empirical findings section

- Table 4 first confirms a standard RPE pattern: CEOs are rewarded for own-firm performance and penalized for peer performance.

- Table 5 then shows that this pattern shifts after the DoJ reform for more exposed firms.
- The local-peer return coefficient becomes more positive, while own-return sensitivity falls.
- Table 6 shows the change is strongest in cash compensation, suggesting board discretion rather than mechanical stock-price effects.
- Figure 2 and Table 7 show the result is robust to alternative treatment and sample definitions.
- Table 8 shows the effect is strongest where collusion is most plausible and where managerial-shareholder incentive wedges are likely larger.
- Figure 3 and Table 9 connect compensation changes to profitability and return comovement.

6.5 Other compensation features section

- The authors do not find large changes in whether firms disclose explicit RPE provisions.
- They do find that explicit peer groups become less oriented toward product-market/local peers after the reform.
- Firms increase strategic and profit-margin performance metrics and reduce sales metrics.
- This is consistent with less emphasis on output expansion and more emphasis on pricing, margins, and discretionary coordination-friendly goals.
- Firms lengthen vesting horizons, consistent with making managers more patient and less likely to undercut local rivals for short-run gains.

6.6 Convicted-cartel section

- The paper uses convicted cartel data as supplementary evidence.
- For firms known to have participated in cartels, CEO pay becomes more positively related to cartel-peer performance after cartel initiation.
- The overlap between cartel peers and compensation benchmarks is larger for general pay benchmarks than for formal RPE benchmarks.
- This suggests firms may avoid explicitly tying RPE awards to cartel peers while still using compensation structures that relate pay to cartel-peer performance.

6.7 Conclusion section

- The paper concludes that weakening antitrust enforcement can change CEO incentives in ways that facilitate collusion.
- The findings create a public-policy tension: stronger alignment between shareholders and managers can reduce consumer welfare when shareholders benefit from anti-competitive conduct.
- The evidence suggests local antitrust enforcement matters, because losing field offices changed both enforcement patterns and corporate compensation design.

7 Short summary

- The paper studies how executive compensation can motivate collusion.
- The empirical shock is the 2013 closure of four DoJ Antitrust Division regional offices.
- The treatment is the increase in distance from firm headquarters to the covering antitrust office.

- The main peer group is local Hoberg-Phillips product-market peers within 200 miles.
- Standard RPE would make CEO pay negatively related to peer returns.
- After the reform, exposed firms make CEO pay more positively related to local peer returns.
- The same firms reduce own-firm pay-performance sensitivity.
- The effect is stronger in cash compensation than equity compensation.
- Robustness tests support the result under alternative treatment definitions, matching, placebo reforms, peer definitions, locality definitions, and fixed effects.
- Heterogeneity is strongest where collusion is easier or where CEO-shareholder incentives diverge more.
- Exposed firms show higher gross margins and greater return comovement with local peers.
- Firms alter explicit contract features: fewer local peers in RPE groups, more strategic/profit-margin metrics, fewer sales metrics, and longer vesting horizons.
- Convicted cartel firms show a similar pattern around actual cartel initiation.
- The paper argues that corporate governance can sometimes help shareholders at the expense of consumers.

8 Conclusion

8.1 Core conclusion

Ha, Ma, and Žaldokas show that executive compensation can be redesigned to make collusion more attractive to managers when antitrust enforcement weakens. The evidence is centered on the 2013 DoJ office closures, which reduced local enforcement intensity for many firms.

8.2 Economic intuition

Managers and shareholders do not necessarily have the same willingness to collude. Managers face prison risk, reputation loss, and career concerns. Shareholders mainly receive the upside from higher margins. When detection risk falls, shareholders have more reason to encourage softer competition. A compensation contract can do this indirectly by rewarding the CEO when local rivals perform well and by lengthening incentives so that the CEO values the long-run gains from coordination.

8.3 Takeaway

The paper adds a cautionary lesson to corporate governance. Aligning managers with shareholders is not always socially benign. When shareholder value rises from reduced competition, better incentive alignment can make firms less competitive and can lower consumer welfare.